

Andrew MM Matheson

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Summary

I am a neuroethologist interested in how animals achieve similar behaviours for survival despite inhabiting vastly different habitats and possessing distinct neural circuit architectures. My current research focuses on the Iberian ribbed newt (*Pleurodeles waltl*), a remarkably plastic species that transitions between aquatic and terrestrial living as an adult — two environments with profoundly different sensory and physical properties. I aim to understand the extent to which nervous systems can be flexible in this species and the morphological, behavioural and genetic modifications newts use to adapt between environments. I plan to leverage the newt's unique plasticity, regenerative capacity and its key evolutionary position as a sister group to all other terrestrial vertebrates to understand the flexibility of the nervous system over evolution. My long term goal is to develop the newt into a full-fledged model for neuroethology and comparative systems neuroscience.

Education

PhD	New York University , Neuroscience and Physiology	New York, NY, USA
	- Thesis - The Neural Circuits that Support Wind-Guided Olfactory Navigation, advised by Katherine I Nagel	Jan 2015 – Jan 2021
BSc	McGill University , Computer Science and Biology	Montreal, QC, Canada
	Researched vocal communication using songbirds in Jon T Sakata's lab	Jan 2011 – Jan 2015

Experience

Tosches Lab, Department of Biological Sciences, Columbia University , Postdoctoral Research Scientist	New York, NY
Investigating neural mechanisms of environmental plasticity and sensory processing in the Iberian ribbed newt.	Jan 2021 – present 5 years 6 months

Publications

Iberian ribbed newts Matheson AMM, Chua NJ, Tosches MA 10.1016/j.cub.2024.11.062 (Current Biology)	Jan 2025
A neural circuit for wind-guided olfactory navigation Matheson AMM, Lanz AJ, Currier TA, Licata AM, Syed MH, Nagel KI 10.1038/s41467-022-32247-7 (Nature Communications)	Jan 2022
A population of central complex neurons promoting airflow-guided orientation in <i>Drosophila</i> Currier TA, Matheson AMM, Nagel KI 10.7554/eLife.61510 (eLife)	Jan 2020
Central neurons encoding wind direction in <i>Drosophila</i> Suver MP, Matheson AMM, Sarkar S, Damiata M, Schoppik D, Nagel KI 10.1016/j.neuron.2019.03.012 (Neuron)	Jan 2019
Cortical control of vocal interaction in neotropical singing mice Okobi DE, Banerjee A, Matheson AMM, Phelps SM, Long MA 10.1126/science.aau9480 (Science)	Jan 2019

Relationship between the sequencing and timing of vocal motor elements in birdsong

Jan 2015

Matheson AMM, Sakata JT

[10.1371/journal.pone.0143203](https://doi.org/10.1371/journal.pone.0143203) (Plos One)

Awards

Doupe Fellowship

Jan 2026

Awarded by the McKnight Endowment Fund for Neuroscience.

Young Investigator Award

Jan 2026

Awarded by the International Society for Neuroethology. Award recognizing early-career achievements in neuroethology.